Parameters Analysis for an Intersection Bus-Pedestrian Collision Warning System

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***Abstract*—Utilizing advanced technologies to guarantee safety protection for pedestrians at intersections has become an important issue because intersections are the places where pedestrians easily conflict with vehicles. The purpose of this study is to propose a conceptual design of an intersection bus-pedestrian collision warning system with appropriate detection and warning parameters for bus drivers approaching an intersection. Based on the design concept, bus drivers' perception-reaction time, emergency deceleration rate of buses, and pedestrian walking speed are defined as the basic parameters. A bus driving simulation is designed and conducted to collect bus drivers' responses to the suddenly crossing pedestrians at unsignalized intersections or signalized intersections with green interval for parameters analysis. Finally, the warning timings for auditory warnings and visual warnings, the locations for vehicle detectors and pedestrian detectors, and the locations for visual warning devices are developed through a further analysis of the experimental results.**

Keywords-pedestrian; intersection; bus; collision warning system; parameter.

I. INTRODUCTION

Intelligent Transportation Systems (ITS) are the systems which apply advanced information, communications, electronics, mechanics, and control technologies to the operations of transportation and traffic systems for solving such problems as traffic accidents, congestion and environmental impacts. Due to the rapid development of information and communications technologies, it has become feasible to apply advanced techniques to solve traditional traffic problems. Recently, the development of ITS in advanced countries has focused on not only the enhancement of service quality for most road users such as vehicle drivers but also the achievement of transportation expectation for road vulnerable individuals such as pedestrians. Thus, using ITS technologies to guarantee safety protection for pedestrians at intersections has become an important issue because intersections are the places where pedestrians easily conflict with vehicles [1-4]. In general, traffic engineers usually design a signal timing plan with pedestrian phases to improve pedestrian crossing safety at an intersection. However, some intersections in rural areas are not available to be signalized because of installation cost and vehicle delay issues. Therefore, protecting pedestrians who are crossing the roads from pedestrian-vehicle collisions at unsignalized intersections is a major concern for traffic engineers and government authorities trying to improve road intersection safety. One way to provide the crossing pedestrians with the higher right of way at unsignalized intersections is making traffic regulations to force drivers to bring their vehicles to a stop to allow pedestrians to cross the road. Another way is to develop an intersection pedestrian collision warning system (IPCWS). IPCWS is one of the pedestrian protection systems integrating advanced detection technologies, warning devices, and collision warning rules to provide timely alert messages to drivers and pedestrians for collision avoidance at intersections. The system can be used to prevent vehicles from colliding with not only regular road crossing pedestrians at unsignalized intersections but also illegal road crossing pedestrians at signalized intersections.

In order to improve intersection safety, much research has been devoted to develop intersection collision warning systems or driving decision support interfaces for collision avoidance. The U.S. Federal Highway Administration (FHWA) [5] developed an intersection collision warning system (ICWS) to enhance driver awareness of traffic situations at intersections. The system provided timely and easily understood warning messages on active signs for vehicles approaching an intersection. A field study showed that the ICWS had a positive effect on vehicle speed reduction. Laberge et al. [6] proposed an infrastructure-based intersection decision support (IDS) system to help drivers make safer gap acceptance decisions at rural intersections. Sign was the major interface of the system for approaching drivers. In the interface design, four major concepts were presented to drivers: hazard sign, countdown sign, variable message sign (VMS), and icon sign. Important information in the sign includes showing the presence of gaps, indicating the size of available gaps, and/or judging the safety of available gaps. Neale et al. [7] designed and evaluated the infrastructure-based warning interfaces and the vehicle-based warning interfaces for a signal-controlled and stop-controlled violation warning system on the Smart Road in Virginia, USA. The evaluation results indicated that vehicle-based warning interfaces could provide effective warnings to the violating drivers. Stubbs et al. [8] and Atev et al. [9] also presented some vision systems for real-time collision warnings at intersections. While most studies have focused on the discussion of collision avoidance between one vehicle and another vehicle, Van Houten et al. [10] evaluated the effectiveness of signs with pedestrian symbol for motorists yielding to pedestrians at crosswalks. They found that one sign installed 50 m in front of the crosswalk would be helpful in reducing conflicts between vehicles and pedestrians. Although many studies have proposed innovations or conducted preliminary tests on ICWS, little information is available on the development of the intersection bus-pedestrian collision warning system.

The accident involving a bus and pedestrians often causes severe injuries to pedestrians due to the high momentum of the bus. Thus it is vital to develop an intersection bus-pedestrian collision warning system for mitigating bus-and-pedestrian collisions at intersections. This paper aims to propose a conceptual design of an intersection bus-pedestrian collision warning system and identify the required parameters. In addition, a bus driving simulation is further conducted to collect and analyze the values of the parameters. Remaining sections of the paper are organized as follows. In Sec. II, a conceptual design of the warning system is briefly described. Then, the experimental design and the parameters analysis are presented in Sec. III. Finally, conclusions are summarized in Sec. IV.

II. CONCEPTUAL DESIGN OF AN INTERSECTION BUS-PEDESTRIAN COLLISION WARNING SYSTEM

Fig. 1 shows the design concept of the collision warning system. In the figure, devices layout and warning timing are determined by the distance between the detection point of the pedestrian and the collision point, and the stopped sight distance of the approaching bus driver. Bus drivers usually drive their buses on the outer lane when they approach an intersection since many bus stops are installed at the near side of the intersection. The VMS for the bus drivers is installed on the median to avoid being obscured by other signs and provide adequate visibility and legibility. When the system initiates and a crossing pedestrian is detected, the warning timing should provide sufficient time for an approaching bus decelerating and stopping in front of the collision point. In other words, the warning timing is the timing which provides a period of time to the driver for receiving the warning messages and stopping the bus. Therefore, the warning timing is given by

$$T_{wd} = T_r + \frac{V_0}{3.6d} \tag{1}$$

where

T_{wd}: warning timing (time interval before the bus is totally stopped) for the bus driver (in seconds);

T_r: perception-reaction time of the bus driver (in seconds);

V_0: approaching speed of the bus (in kilometers per hour);

d: emergency deceleration rate of the bus (in meters per square second).

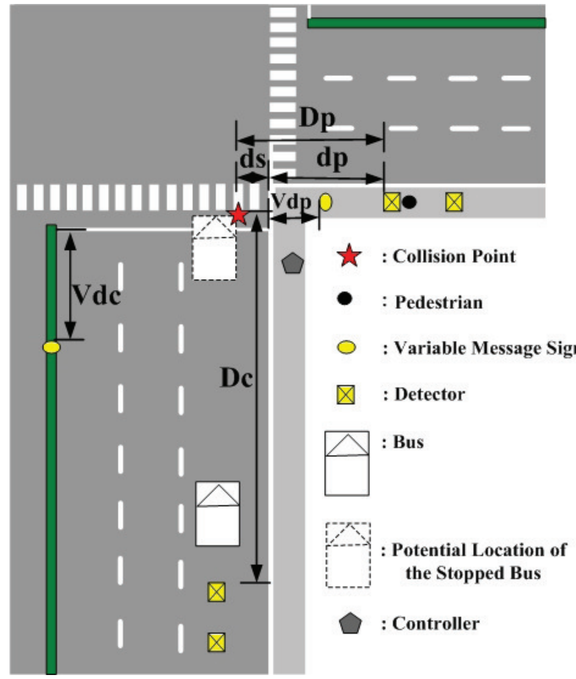

Figure 1. The layout of the conceptual intersection bus-pedestrian collision warning system.

According to the required warning timing in (1), the shortest distance between the pedestrian detector and the collision point (Dp) could be given by

$$\text{Dp} = \text{ds} + \text{dp} = (T_{wd} + T_{b1}) \times V_p \tag{2}$$

$$\text{dp} = (T_{wd} + T_{b1}) \times V_p - \text{ds} \tag{3}$$

where

ds: distance between the collision point and the curb (in meters)

dp: the shortest distance between the pedestrian detector and the curb (in meters);

T_{wd}: warning timing for the bus driver (in seconds);

T_{b1}: buffer time from the detection of pedestrian to the warning message initiation (in seconds);

V_p: walking speed of the pedestrian (in meters per second).

In practice, ds could be set to zero, and Dp is equal to dp. Equation (3) could be rewritten by

$$\text{dp} = (T_{wd} + T_{b1}) \times V_p \tag{4}$$

Equation (4) means that the detection range of crossing pedestrians should be at least dp for providing sufficient warning timing to bus drivers. The shortest distance between the vehicle speed detector and the collision point (Dc) is given by

$$Dc = 0.278\, V_0 \times (T_r + T_{b2}) + \frac{V_0^2}{2\times(3.6)^2 \times d}$$
$$= 0.278\, V_0 \times (T_r + T_{b2}) + \frac{V_0^2}{25.92\times d} \quad (5)$$

where

Dc: the shortest distance between the vehicle speed detector and the collision point (in meters);

T_r: perception-reaction time of the bus driver (in seconds);

T_{b2}: buffer time for vehicle speed calculation (in seconds);

V_0: approaching speed of the bus (in kilometers per hour);

d: emergency deceleration rate of the bus (in meters per square second).

Sometimes the information of the oncoming bus should be provided to the pedestrians if the bus does not decrease its speed. The warning timing for the pedestrians is given by

$$T_{wp} = (3.6\text{Dc}) / V_0 \quad (6)$$

where

T_{wp}: warning timing for the pedestrians (in seconds);

Dc: the shortest distance between the vehicle speed detector and the collision point (in meters);

V_0: approaching speed of the bus (in kilometers per hour).

In addition to the warning timings and the detector locations, the locations of variable message signs (Vdc and Vdp) should be determined in Fig. 1 if the warning messages are presented on variable message signs. In general, Vdc is smaller than Dc and Vdp is smaller than dp. Vdc and Vdp could be determined by the stopping sight distance of bus drivers and the stopping sight distance of pedestrians, respectively.

A simple flowchart of the warning logic is shown in Fig. 2. When the warning system initiates, the vehicle detection system is used to detect the approaching buses. The controller calculates the bus speed by the detector data and computes the warning timing for pedestrians by (6) if an approaching bus is detected. Then the computed warning timing is displayed on a VMS for pedestrians in a countdown form. If a crossing pedestrian is detected by the pedestrian detector after a bus has been detected, the controller will provide warning messages to the bus driver for decelerating the bus.

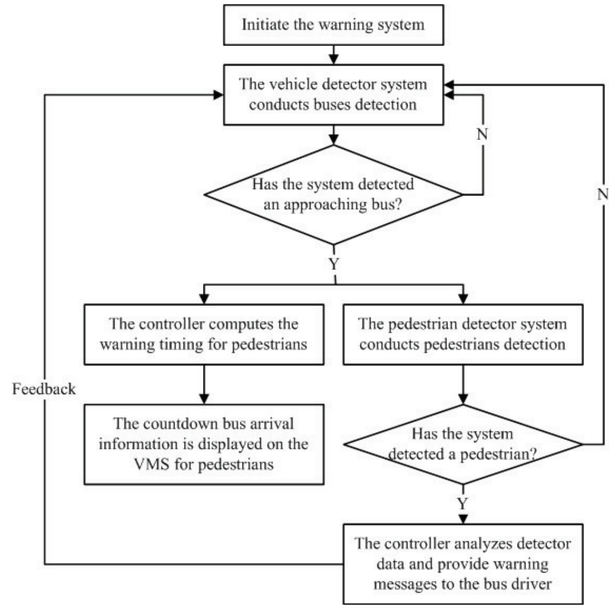

Figure 2. The warning logic of the system.

III. EXPERIMENTAL DESIGN AND PARAMETERS ANALYSIS

According to the conceptual design of the intersection bus-pedestrian collision warning system, perception-reaction time of the bus driver, emergency deceleration rate of the bus, and pedestrian walking speed are the basic parameters of the system. This work conducts a bus driving simulation to collect bus drivers' perception-reaction time and emergency deceleration rate for parameters analysis. Moreover, the mean walking speeds for elderly pedestrians, adult pedestrians and children at intersections are 0.88, 1.29 and 1.16 m/s, and the 15^{th} percentile walking speeds for elderly pedestrians, adult pedestrians and children are 0.72, 1.10 and 1.00 m/s from a field study. The 15^{th} percentile walking speeds are considered in the driving simulation.

A. Experimental Design

The bus driving simulator used in the experiment is the CHU-DS [11]. Fig. 3 shows the designed scenario of the driving experiment. This work uses the EON and 3D MAX packages to create a 3-D simulated driving environment. All driving vehicles except the host vehicle (bus) are small passenger cars. The bus (host vehicle) is stopped in the outer lane upon simulation initiation. Each roadway includes three, 3.75 meter-width lanes and one, 2 meter-width sidewalk. A 2 meter-width median separates different roadways. The road geometry is level terrain (0%) and the weather is sunny and daytime. The speed limit of vehicles is 50 km/h, and the traffic level-of-service (LOS) in each lane (except the lane in which the bus is driving) is B.

(a)

(b)

Figure 3. The driving scenario of the experiment. (a) The scene without crossing pedestrians. (b) The scene with crossing pedestrians.

While conducting the experiment, eighteen events which consist of three experimental factors with two or three levels (see Table I) are initiated randomly for each subject to respond. For example, an event which consists of usignalized intersection, children and voice warning means that a voice warning initiates when the subject is approaching to a crossing child at an unsignalized intersection. Each event is initiated when the subject arrives about 70 m in front of the edge of the crosswalk. The VMS is installed at 60 m in front of the edge of the crosswalk (see Fig. 4). Each subject should conduct the whole experiment twice and face each event twice during the driving simulation.

TABLE I. THE EXPERIMENTAL FACTORS AND LEVELS

Intersection Types	Pedestrian Types	Warning Types
1.Unsignalized intersection 2.Signalized intersection with green interval	1.Children 2.Adult pedestrians 3.Elderly pedestrians	1.Voice warning 2.VMS display 3.No warning

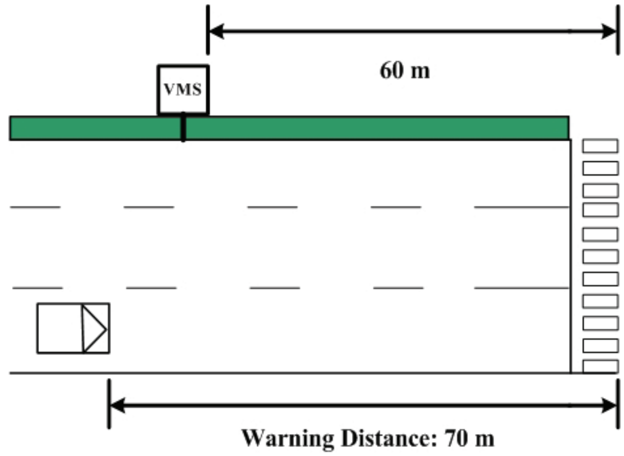

Figure 4. The warning distance and VMS distance in the driving scenario.

This work recruits eighteen bus drivers working with a freeway bus company in Taiwan. All are licensed male drivers. Driving experiences of the subjects ranged from 3 to 30 years, with an average of 15 years.

The measures include accident happenings, perception-reaction time, and emergency deceleration rate. The accident happenings are defined as "the number of bus-pedestrian collisions in the experiment". The perception-reaction time is defined as "the time period from the warning messages initiation to the deceleration initiation of the bus (for voice warning type and VMS display type)" or "the time period from the appearance of a pedestrian to the deceleration initiation of the bus (for no warning)". The emergency deceleration rate is defined as "the average change rate of the bus speed while braking". It is the average rate defined as the bus speed before braking divided by the total brake time.

B. Experimental Results

After the experiment, this work analyzes accident happenings, perception-reaction time, and emergency deceleration rate from the driving simulation. The analysis results are described as follows.

1) Accident happenings

Two bus-pedestrian accidents happened while a pedestrian was crossing an unsignalized intersection without warning message devices. It indicates that providing appropriate warning messages to bus drivers who are approaching to an intersection can avoid bus-pedestrian collisions effectively.

2) Perception-reaction time

Table II shows the statistics of perception-reaction time for the subjects responding to various events. We can find that the average perception-reaction time with warning messages is higher than that without warning messages. It implies that warning messages provide drivers with sufficient time to stop the bus before accidents happen. The shorter perception-reaction time without warning messages also implies that bus drivers can quickly react to an emergency situation.

TABLE II. THE STATISTICS OF PERCEPTION-REACTION TIME

Intersection Types	Pedestrian Types	Warning Types	Sample Size	Perception-reaction Time, seconds	
				Mean	Standard Deviation
Unsignalized intersection	Children	Voice warning	36	2.23	0.66
		VMS display	34	2.19	0.70
		No warning	34	1.76	0.54
	Adult pedestrians	Voice warning	36	2.07	0.66
		VMS display	35	2.13	0.56
		No warning	35	1.66	0.61
	Elderly pedestrians	Voice warning	36	2.21	0.54
		VMS display	35	2.13	0.57
		No warning	36	1.72	0.61
Signalized intersection with green interval	Children	Voice warning	34	2.27	0.64
		VMS display	34	2.23	0.57
		No warning	35	1.72	0.88
	Adult pedestrians	Voice warning	35	2.14	0.65
		VMS display	34	2.23	0.64
		No warning	36	1.61	0.69
	Elderly pedestrians	Voice warning	36	2.19	0.64
		VMS display	35	2.32	0.70
		No warning	36	1.60	0.62

3) Emergency deceleration rate

The statistics of emergency deceleration rate for the subjects responding to various events are summarized in Table III. We can find that in most cases the average emergency deceleration rates without warning messages are higher than those with warning messages. It indicates that bus drivers have to brake suddenly to react to an emergency situation without early warnings even though they can quickly respond to it.

The experimental results of perception-reaction time and emergency deceleration rate are further analyzed by factorial analysis of variance (ANOVA) to compare the significance of difference of the measures. The analysis results show that the difference of the mean perception-reaction time and the difference of the mean emergency deceleration rate among warning types are significant, with $\alpha = 0.05$. The average perception-reaction time with warning messages is significantly higher than that without warning messages, and the emergency deceleration rates without warning messages are significantly higher than those with voice warnings, with $\alpha = 0.05$.

TABLE III. THE STATISTICS OF EMERGENCY DECELERATION RATE

Intersection Types	Pedestrian Types	Warning Types	Sample Size	Emergency Deceleration Rate, m/sec^2	
				Mean	Standard Deviation
Unsignalized intersection	Children	Voice warning	36	3.15	1.23
		VMS display	36	3.26	1.27
		No warning	35	3.46	1.29
	Adult pedestrians	Voice warning	36	3.17	1.14
		VMS display	36	2.99	1.11
		No warning	35	3.47	1.09
	Elderly pedestrians	Voice warning	36	3.18	1.38
		VMS display	36	2.97	1.09
		No warning	36	3.56	1.44
Signalized intersection with green interval	Children	Voice warning	34	3.01	1.19
		VMS display	34	3.35	1.36
		No warning	35	3.10	1.18
	Adult pedestrians	Voice warning	35	2.98	1.02
		VMS display	34	3.41	1.28
		No warning	36	3.25	1.19
	Elderly pedestrians	Voice warning	36	2.77	1.20
		VMS display	35	3.03	1.31
		No warning	36	3.29	1.39

C. Parameters Analysis

The values of parameters, including the warning timing for the bus driver (T_{wd}), the shortest distance between the pedestrian detector and the curb (dp), the shortest distance between the vehicle speed detector and the collision point (Dc), the warning timing for the pedestrians (T_{wp}), and the locations of variable message signs (Vdc and Vdp), can be developed through the further analysis of the perception-reaction time, the emergency deceleration rate and other parameters from the simulation experiment. The analysis results are described as follows.

1) The warning timing for the bus driver (T_{wd})

During the experiment, the locations of the stopped buses are about 11-13 m in front of the edge of the crosswalk if the buses did not collide with the crossing pedestrians, thus the (1) is modified by

$$T_{wd} = T_r + \frac{V_0}{3.6d} - \beta \tag{7}$$

The β is an adjustment factor which is computed as the distance between the stopped bus and the stop line (the distance between the stopped bus and the edge of the crosswalk minus 5.5 m) divided by the approaching speed of the bus (V_0). The approaching speed of the bus is assumed as 50 km/h. Table IV shows the mean warning timing for bus drivers avoiding colliding with various types of pedestrians at different intersection types with warning devices. Because the difference of the mean perception-reaction time and the difference of the mean emergency deceleration rate are not significant in intersection types and pedestrian types, we develop generalized warning timings for the voice warning and the VMS display as 6.28 s and 6.07 s, respectively.

TABLE IV. THE MEAN WARNING TIMING FOR THE BUS DRIVERS

Intersection Types	Pedestrian Types	T_{wd} for voice warning, seconds	T_{wd} for VMS display, seconds
Unsignalized intersection	Children	6.15	5.96
	Adult pedestrians	5.96	6.28
	Elderly pedestrians	6.11	6.27
Signalized intersection with green interval	Children	6.40	5.70
	Adult pedestrians	6.35	5.85
	Elderly pedestrians	6.77	6.46
Generalized Value		6.28	6.07

2) The shortest distance between the pedestrian detector and the curb (dp)

The location of the pedestrian detector is determined by (4) and the highest pedestrian speed (1.11 m/s for adult pedestrians). The buffer time from the detection of pedestrian to the warning message initiation (T_{b1}) is determined by the detection technology. We assume that the value is 0.1 s, and then the shortest distance between the pedestrian detector and the curb is computed as 7.1 m.

3) The shortest distance between the vehicle speed detector and the collision point (Dc)

Considering the locations of the stopped buses during the experiment, we modify (5) by

$$\mathrm{Dc} = 0.278\ V_0 \times (T_r - \beta + T_{b2}) + \frac{V_0^2}{25.92 \times d} \quad (8)$$

The buffer time for the calculation of vehicle speed (T_{b2}) is assumed as 0.1 s. The computed value for the shortest distance between the vehicle speed detector and the collision point is 61 m.

4) The warning timing for the pedestrians (T_{wp})

By (6) and the computed Dc, the warning timing for pedestrians is computed as 4.39 s.

5) The locations of variable message signs (Vdc and Vdp)

Based on the experimental results, we adjust the location of the VMS for bus drivers and recommend that the distance between the VMS and the stop line is 50 m. The distance between the VMS for pedestrians and the curb (Vdp) is determined by the perception-reaction distance (perception-reaction time × walking speed) of pedestrians. We use the design value for the perception-reaction time (2.5 s) and the mean walking speed for adult pedestrians (1.29 m/s) to compute the distance. Finally, the suggested value is 3 m.

IV. CONCLUSIONS AND DISCUSSION

According to the traffic accident statistics in some prior studies, high percentage of fatal accidents involves pedestrians crossing the street at intersections. It also implies that intersections are the high-risk places for pedestrians. With the rapid development of information and communications technologies, using advanced technologies to improve pedestrian safety at intersections has become more feasible. Based on the concept of the intersection collision warning system (ICWS), this study has attempted to propose an innovation design in the collision warning system for both bus drivers and pedestrians at intersections. Understanding the importance of parameters for an adequate warning system, we conduct a bus driving simulation experiment and analyze experimental results to develop the values for the system parameters, including the warning timing for the bus driver, the shortest distance between the pedestrian detector and the curb, the shortest distance between the vehicle speed detector and the collision point, the warning timing for the pedestrians, and the locations of variable message signs.

The experimental results also reflect some characteristics of bus driving behavior at intersections such as quickly reacting to emergency situations but braking suddenly in the emergency situations without early warnings. Sometimes the buses might collide with pedestrians in the emergency situations. Thus, it is helpful to provide appropriate warning messages to bus drivers who are approaching to the pedestrians crossing the street at an intersection for avoiding bus-pedestrian collisions effectively. The conceptual design and parameters developed in this study should provide a useful base in the development of an intersection bus-pedestrian collision warning system.

For the layout of the warning system, we believe that the locations of variable message signs will cause impacts on bus drivers and pedestrians. It will be necessary to conduct more detailed simulations to validate the concept. Moreover, the detection and telecommunication techniques should be further analyzed to enhance the capabilities of the system.

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